

TOP 10

Open Source Databases

A humongous amount of data is being generated across the globe each day, thanks to technologies like the Internet of Things and artificial intelligence. This article briefly introduces some of the best open source tools that can help to manage this data.

Traditionally, databases have been managed by proprietary tools offered by Oracle, IBM, Microsoft, and various other smaller organisations. Lately, however, and particularly for new ventures, open source databases have become significantly popular.

Latest trends related to database management

In 2020, database management went through a number of advancements, and most individuals made use of social databases or information distribution centres. The latter coordinate value-based and operational information, and give important business intelligence (BI) through reports and analytics.

To meet the objectives of database management, organisations need to:

- Optimise data warehouses
- Leverage cloud (distributed) computing
- Include graph databases
- Consider multidimensional database management systems (MDBMS)
- Provide AI frameworks
- Augment data management

Multiple databases

Practically everyone is running multiple databases, and multiple open source databases at that. When it comes to using different data storage technologies to handle different needs, a majority are using both relational and other databases, while a smaller share of users is running non-relational databases. Today, many non-relational databases (or data management platforms, if we want to include solutions

like Hadoop, for example) support SQL.

Open source databases in the cloud

If you are using cloud management software or a cloud broker service, it's imperative to have the option to store your information in the kind of database you require.

Some of the best cloud databases are:

- Amazon relational database
- Microsoft Azure SQL database
- Oracle database
- IBM Db2 on cloud
- Google Cloud SQL

Figure 1 lists the best free and open source database management systems and their type.

S. No.	Database Name	Type
1	PostgreSQL	Object-Relational Database Management System
2	MariaDB	Relational Database Management System
3	CockroachDB	Relational Database Management System
4	Neo4j	Graph Database Management System
5	CouchDB	NoSQL Database Management System
6	RethinkDB	Distributed Document-oriented Database Management
7	Redis	NoSQL Database Management System
8	SQLite	Relational Database Management System
9	Cassandra	NoSQL Database Management System
10	Timescale	NoSQL Database Management System

Figure 1: Top 10 free and open source databases and their type